



PROFILE

Ambitious and driven Mechanical Engineering graduate with a strong academic record and a wide range of hands-on experience. Known for a collaborative approach, a genuine passion for engineering, and the ability to adapt quickly across technical and people-facing environments. Eager to grow within a forward-thinking organisation and make a meaningful contribution from day one.

CONTACT

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TECHNICAL SKILLS

- Simulation & Analysis
 - FEA & CFD (ANSYS, SolidWorks)
 - Structural, thermal & flow analysis
- Programming & Control
 - MATLAB & Simulink
 - Python & mathematical modelling
- PID control & feedback systems
- Design & Manufacturing
 - CAD & CNC programming
 - SolidWorks (3D modelling)
- Energy & Thermofluids
 - Thermodynamics & heat transfer
 - Green hydrogen & electrolysis
 - Fluid mechanics & system design
- Digital & Communication
 - Adobe Illustrator & Photoshop
 - Social media & content creation
- Microsoft Office Suite

PROFESSIONAL

- Problem-solving & critical thinking
- Project planning & delivery
- Team leadership & collaboration
- Technical writing & reporting
- Teaching, tutoring & mentoring
- Adaptability & fast learning

LANGUAGES

- English (UKVI IELTS 8.0)
- Sinhala (Native)

JOSHUA FERNANDO

Mechanical Engineer • BEng (Hons)

WORK EXPERIENCE

– Engineering Apprentice — U.S.S Engineering (Huawei Partnership)

Colombo, Sri Lanka | Aug 2018 – Apr 2019

- Supported deployment of fibre-optic infrastructure across rural regions; conducted site assessments to optimise cable routing and network layout
- Ensured QA compliance with international telecom standards, strengthening technical communication and field engineering skills

– STEM Tutor & Ambassador — CTC (Creative Tuition Collective)

Bristol, United Kingdom | Aug 2023 – Mar 2024

- Delivered Engineering and Mathematics tuition to students aged 17–22 across online and in-person settings, improving student performance
- Represented CTC as STEM Ambassador, conducting outreach sessions in Bristol schools to promote engineering careers

– Activity Instructor — Bristol Activity Centre

Bristol, United Kingdom | Apr 2024 – Jul 2025

- Led structured activity sessions for diverse groups, managing safety protocols, risk assessments, and participant welfare consistently

– Social Media Assistant / Graphic Designer — MOD'Art University

Colombo, Sri Lanka | Oct 2020 – Aug 2021

- Produced graphics, promotional content, and managed social platforms; applied Adobe Illustrator & Photoshop for photo editing and visual design

– Private Tutor — Sri Lanka

Aug 2019 – Nov 2022

- Delivered A-Level Physics, Mathematics, and Chemistry tuition online and in person, consistently achieving strong student outcomes

EDUCATION

– BEng (Hons) Mechanical Engineering — Grade: 2:1

University of the West of England, Bristol, United Kingdom | Nov 2025

- Ranked 1st in cohort for Engineering Mathematics (98%); 2nd overall in Year 1. Awarded merit-based scholarship for exceptional academic performance

Modules: Thermodynamics | Fluid Mechanics | Applied Thermo-fluids | Dynamics | Vibrational Dynamics | Control Engineering | Structural Mechanics | Materials & Structures | Advanced Manufacturing | System Design

– Cambridge A-Levels — Physics | Mathematics | Chemistry

Lyceum International School, Sri Lanka | 2017 – 2019

KEY PROJECTS

– Dissertation: Solar-Powered Green Hydrogen Generation

- Designed and modelled AWE & PEM electrolysis systems integrated with solar PV; used MATLAB & FEA to optimise system performance and electrode geometry
- Developed a 3D-printed gyroid electrode achieving measurable improvement in hydrogen output efficiency; analysed energy recovery via HICEs and fuel cells

– Automated Golf Placement Robot (AL5D Robotic Arm)

- Engineered a PID-controlled robotic arm system with real-time sensor feedback; implemented control algorithms achieving precise, repeatable ball placement

– Sustainable Transport — DIY Electric Bike Conversion Kit

- Led multidisciplinary team to design an EV conversion kit and public charging network; conducted full feasibility analysis and energy demand modelling

References available on request